

6

Tate's Thesis

Stephen S. Kudla*

Tate's thesis, *Fourier Analysis in Number Fields and Hecke's Zeta-Functions* (Princeton, 1950) first appeared in print as Chapter XV of the conference proceedings *Algebraic Number Theory*, edited by Cassels and Frölich (Thompson Book Company, Washington, DC, 1967). In it, Tate provides an elegant and unified treatment of the analytic continuation and functional equation of the L -functions attached by Hecke to his Größencharaktere in his pair of papers [7]. The power of the methods of abelian harmonic analysis in the setting of Chevalley's adèles/idèles provided a remarkable advance over the classical techniques used by Hecke.¹ A sketch of the analogous interpretation of classical automorphic forms in terms of *nonabelian* harmonic analysis—automorphic representations—is given in the next chapter. The development of the theory of automorphic representations for arbitrary reductive groups G is one of the major achievements of mathematics of the later part of the twentieth century. And, of course, this development is still under way, as is evident from the later chapters [6]. In hindsight, Tate's work may be viewed as giving the theory of automorphic representations and L -functions of the simplest connected reductive group $G = \mathrm{GL}(1)$, and so it remains a fundamental reference and starting point for anyone interested in the modern theory of automorphic representations.

These articles have two main goals. The first is to give a unified treatment, following Tate, of the analytic properties, analytic continuation, functional equation, etc., of the L -function $L(s, \chi)$ attached to any Hecke character χ for any number field. The second is to introduce the point of view and techniques of representation theory of adèle groups in the simplest case. For example, the progression from local results to global results and the role played by uniqueness theorems for eigendistributions is rather typical in this business. The following chapters provide only a sketch and the reader should consult more extended sources for details.

References

Of course, the best reference is Tate's thesis itself. In addition, there are many other expositions, including Weil [13, Chapter VII], Bump [2, Chapter 3.1], [5], [9], and

*This chapter is a slightly expanded version of two lectures on Tate's thesis given at the school on Automorphic Forms, L-Functions, and Number Theory at Hebrew University in Jerusalem in March of 2001.

¹Tate gives some interesting historical comments in the thesis and at the end of [3, Chapter XV].

the recent book of Valenza and Ramakrishnan [11]. In the present condensed survey, I more or less follow the approach sketched in Weil's 1966 Bourbaki talk *Fonction zeta et distributions* [12, 14], which places a greater emphasis on the equivariant distributions which are only implicit in Tate's original work. Weil's commentary on [14, Volume III, p. 448] is also of interest.

1 Adèles, idèles, and Dirichlet characters

As a motivation for the role played by the idèles and quasicharacters, consider the following description of classical Dirichlet characters.

For a positive integer N , a classical Dirichlet character $\underline{\chi}_N : \mathbb{Z} \rightarrow \mathbb{C}$ modulo N is a function obtained from a character $\chi_N : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ by first extending by 0 on $\mathbb{Z}/N\mathbb{Z}$ and then pulling back to $\mathbb{Z}$. We will call χ_N a Dirichlet character. If $N \mid M$, then χ_N defines a character $\chi_M = \chi_N \circ \text{pr}_{M,N}$ of $(\mathbb{Z}/M\mathbb{Z})^\times$, by pulling back under the projection

$$\text{pr}_{M,N} : (\mathbb{Z}/M\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/N\mathbb{Z})^\times, \quad N \mid M,$$

and there is an associated classical Dirichlet character $\underline{\chi}_M$ modulo M as well. The inverse limit of the system $\{(\mathbb{Z}/N\mathbb{Z})^\times, \text{pr}_{M,N}\}$ is the compact, totally disconnected, topological group²

$$\widehat{\mathbb{Z}}^\times := \varprojlim_N (\mathbb{Z}/N\mathbb{Z})^\times,$$

and every Dirichlet character χ_N can be viewed as a *continuous* character χ of $\widehat{\mathbb{Z}}^\times$. The *conductor* of χ is the smallest N_0 for which χ is trivial on the kernel of the projection $\text{pr}_{N_0} : \widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/N_0\mathbb{Z})^\times$. Then $N_0 \mid N$ and χ is the pullback of a unique Dirichlet character χ_{N_0} of $(\mathbb{Z}/N_0\mathbb{Z})^\times$. The collection of classical Dirichlet characters $\underline{\chi}_M$ for $M \mid N_0$ all correspond to χ , where $\underline{\chi}_{N_0}$ is the unique *primitive* classical Dirichlet character, and the others are *imprimitive*. An analogous phenomena takes place in the dictionary between classical holomorphic modular forms and automorphic representations of $\text{GL}(2)$, where the normalized newforms play the role of the primitive classical Dirichlet characters.

²Recall that this is, by definition,

$$\widehat{\mathbb{Z}}^\times = \{ (x_N) \in \prod_N (\mathbb{Z}/N\mathbb{Z})^\times \mid \text{pr}_{M,N}(x_M) = x_N \}.$$

The product of the finite groups $(\mathbb{Z}/N\mathbb{Z})^\times$ is compact and hence so is the closed subgroup $\widehat{\mathbb{Z}}^\times$. The projection maps

$$\text{pr}_N : \widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$$

are all surjective and continuous.

Recall that, for a fixed prime p ,

$$\begin{aligned}\mathbb{Z}_p &:= \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} \\ &= \{ \alpha = a_0 + a_1p + a_2p^2 + \cdots \mid 0 \leq a_i \leq p-1 \}, \\ \mathbb{Z}_p^\times &= \{ \alpha \in \mathbb{Z}_p \mid a_0 \neq 0 \}, \\ \mathbb{Q}_p &:= \{ \alpha = \sum_i a_i p^i \}\end{aligned}$$

and so on.

Recall that the adèle ring $\mathbb{A}$ of $\mathbb{Q}$ is the restricted product³

$$\mathbb{A} = \mathbb{R} \times \prod'_p \mathbb{Q}_p$$

with respect to the subrings $\mathbb{Z}_p$. Also $\mathbb{Q} \hookrightarrow \mathbb{A}$, $\alpha \mapsto (\dots, \alpha, \alpha, \alpha, \dots)$, embedded diagonally, is a discrete subring. Similarly, the idèle group is the restricted product

$$\mathbb{A}^\times = \prod'_v \mathbb{Q}_v^\times$$

with respect to the subgroups $\mathbb{Z}_p^\times$, there is a diagonal embedding $\mathbb{Q}^\times \hookrightarrow \mathbb{A}^\times$, and there is a decomposition

$$\mathbb{A}^\times \simeq \mathbb{Q}^\times \times \mathbb{R}_+^\times \times \widehat{\mathbb{Z}}^\times, \quad x = \alpha t u. \quad (1.1)$$

From this last decomposition, we see that any Dirichlet character χ defines a continuous character

$$\omega : \mathbb{Q}^\times \backslash \mathbb{A}^\times \longrightarrow \mathbb{C}^\times, \quad \omega(x) = \omega(\alpha t u) = \chi(u).$$

The most general such *quasicharacter*⁴ has the form $\omega \cdot |\cdot|^s$, for $s \in \mathbb{C}$, where $|x| = |\alpha t u| = t$. Thus, the Dirichlet characters are precisely the quasicharacters of finite order.

Note that, since

$$(\mathbb{Z}/N\mathbb{Z})^\times = \prod_p (\mathbb{Z}/p^n\mathbb{Z})^\times, \quad n = n_p = \text{ord}_p(N),$$

any Dirichlet character χ_N has a factorization

$$\chi_N = \otimes_p (\chi_N)_p,$$

where $(\chi_N)_p$ is a character of $(\mathbb{Z}/p^n\mathbb{Z})^\times$. This gives rise to a *factorization*

$$\chi = \otimes_p \chi_p$$

³Almost all components of an element (x_p) must lie in the subring $\mathbb{Z}_p$.

⁴A continuous homomorphism to $\mathbb{C}^\times$ is called a character if it takes values in $\mathbb{C}^1$, i.e., if it is unitary.

of the corresponding character χ of $\widehat{\mathbb{Z}}^\times$, where each χ_p is the character of $\widehat{\mathbb{Z}}_p^\times$ associated to $(\chi_N)_p$. Note that, for each place v , i.e., for each $\mathbb{Q}_p$ and for $\mathbb{Q}_\infty = \mathbb{R}$, we have an inclusion

$$\mathbb{Q}_v^\times \hookrightarrow \mathbb{Q}^\times \backslash \mathbb{A}^\times \quad x_v \mapsto (\dots, 1, 1, x_v, 1, 1, \dots).$$

Thus, any quasicharacter ω determines quasicharacters ω_v of each $\mathbb{Q}_v^\times$, and there is a factorization

$$\omega = \otimes_v \omega_v \tag{1.2}$$

compatible with the previous one.

Exercise: (i) Check that an *odd* classical Dirichlet character $\underline{\chi} \bmod N$ (i.e., $\underline{\chi}(-a) = -\underline{\chi}(a)$) yields a character ω such that $\omega_\infty(x) = \text{sgn}(x) = x/|x|$.

(ii) For ω associated to a classical Dirichlet character $\underline{\chi}_N$, modulo N , determine $\omega_p(p)$ for primes $p \nmid N$ and for primes $p \mid N$.

2 L -functions for quasicharacters

For k a general number field (or function field over a finite field),⁵ there is a similar picture. We will use the following notation:

$$\begin{aligned} \mathcal{O} &= \text{ring of integers in } k, \\ \mathcal{O}_v &= \text{its completion at a nonarchimedean place } v (= \text{a dvr}), \\ \mathcal{P}_v &= \text{the maximal ideal of } \mathcal{O}_v, \\ \varpi_v &= \text{a generator of } \mathcal{P}_v, \\ \mathbb{F}_v &= \mathcal{O}_v/\mathcal{P}_v = \text{the residue field at } v, \\ q_v &= |\mathcal{O}_v/\mathcal{P}_v| = \text{its order}, \\ \text{ord}_v : k_v &\longrightarrow \mathbb{Z} \cup \{\infty\}, \quad \text{the valuation, with } \text{ord}_v(\varpi_v) = 1, \\ |x|_v &:= q_v^{-\text{ord}_v(x)}, \quad \text{the normalized absolute value,} \\ \mathcal{O}_v^\times &= \text{the group of units,} \end{aligned}$$

and so, $\mathcal{O}_v^\times \times \mathbb{Z} \xrightarrow{\sim} k_v^\times$, $(u, n) \mapsto u\varpi_v^n$. For an archimedean place v with $k_v \simeq \mathbb{R}$ (resp., $k_v \simeq \mathbb{C}$), let $|x|_v$ denote the usual absolute value (resp., the square of the usual absolute value).

Then we have restricted products, the adèles,

$$\mathbb{A} = \prod'_v k_v \quad \text{and the idèles,} \quad \mathbb{A}^\times = \prod'_v k_v^\times,$$

with diagonal embeddings $k \hookrightarrow \mathbb{A}$ and $k^\times \hookrightarrow \mathbb{A}^\times$. Here restricted means that for $x = (\dots, x_v, \dots)$ in $\mathbb{A}$ (resp., $\mathbb{A}^\times$), almost all components x_v lie in $\mathcal{O}_v$ (resp.,

⁵We will exclude this case, simply to avoid the extra side comments it would require.

$\mathcal{O}_v^\times$). There is an absolute value

$$|\cdot|_{\mathbb{A}} : \mathbb{A}^\times \longrightarrow \mathbb{R}_+^\times, \quad |x| = \prod_v |x_v|_v,$$

which is trivial on $k^\times$ (the product formula). The decomposition (1.1) becomes more complicated due to (i) global units and (ii) the nontriviality of the ideal class group. These are precisely the difficulties which Hecke had to work hard to overcome and which, by contrast, the adèlic formalism handles so beautifully. In any case, there are inclusions

$$i_v : k_v^\times \hookrightarrow k^\times \backslash \mathbb{A}^\times, \quad x_v \mapsto (\dots, 1, 1, x_v, 1, 1, \dots),$$

as before.

Definition 2.1. A *quasicharacter* or Hecke character is a continuous complex character $\omega : \mathbb{A}^\times \rightarrow \mathbb{C}^\times$, trivial on $k^\times$.

An explanation of the relation between this definition and the classical notion of Hecke character described in Kowalski's article [8] is given in Section 5 below.

There is a factorization

$$\omega = \otimes_v \omega_v,$$

where $\omega_v = \omega \circ i_v$.

Definition 2.2.

(i) For a nonarchimedean place v , a quasicharacter $\omega_v : k_v^\times \rightarrow \mathbb{C}^\times$ is *unramified* if it is trivial on $\mathcal{O}_v^\times$, and hence can be written in the form

$$\omega_v(x) = t_v^{\text{ord}_v(x)}$$

for $t_v = t_v(\omega_v) \in \mathbb{C}^\times$.

(ii) A quasicharacter ω of $k^\times \backslash \mathbb{A}^\times$ is unramified at v if its local component $\omega_v = \omega \circ i_v$ at v is unramified.

For any ω , there is a finite set of places $S = S(\omega)$, including all archimedean places⁶ such that ω_v is unramified for all $v \notin S$. For example, in the case $k = \mathbb{Q}$ discussed in the previous section, if ω is associated to a primitive classical Dirichlet character $\underline{\chi}_N$ modulo N , then $S(\omega) = \{\infty\} \cup \{p \mid p \mid N\}$. Thus, associated to ω is a collection of complex numbers $\{t_v(\omega)\}_{v \notin S}$. For example, for $s \in \mathbb{C}$, the quasicharacter $\omega_s(x) = |x|_{\mathbb{A}}^s$ is everywhere unramified and determines the set $\{q_v^{-s}\}$.

Definition 2.3. The (partial) *L-function* associated to a quasicharacter ω is the Euler product

$$L^S(s, \omega) = \prod_{v \notin S} (1 - t_v(\omega) q_v^{-s})^{-1}, \quad (\text{EP})$$

⁶Where the notion of “unramified” is not defined.

where $S \supset S(\omega)$. The factors

$$L_v(s, \omega_v) := (1 - t_v(\omega) q_v^{-s})^{-1}$$

are the *local L-factors* associated to the ω_v 's.

Of course,

$$L^S(s, \omega) = L^S(0, \omega\omega_s),$$

so we could dispense with s in the notation. Usually, we will prefer to keep s and to assume that ω is a character instead. In this case, the Euler product is absolutely convergent in the half plane $\operatorname{Re}(s) > 1$.

The main goals are to prove the analytic properties of these functions, i.e.,

1. to “complete” the partial L -function $L^S(s, \omega)$ by including additional local L -factors for the primes $v \in S$, for example, for the archimedean places, and
2. to prove the meromorphic analytic continuation and functional equation of the completed L -function.

These goals will be achieved by interpreting the local L -factors (resp., the global Euler product) as constants of proportionality between two naturally constructed basis elements of a one-dimensional complex vector space of *distributions*!

3 Local theory

Eigendistributions

We fix a place v of k and, to streamline notation, we write $F = k_v$. Let dx (resp., $d^\times x$) be a Haar measure on F (resp., $F^\times$). Observe that, for $a \in F^\times$,

$$d(ax) = |a| dx$$

for the normalized absolute value, as defined above. Note that one then has $d^\times x = \mu |x|^{-1} dx$ for some positive constant μ . Also, for $s \in \mathbb{C}$, let

$$\omega_s(x) = |x|^s$$

be the associated character. Of course, ω_0 is the trivial character. If v is nonarchimedean, we write $\mathcal{O} = \mathcal{O}_v$, $\mathcal{P} = \mathcal{P}_v$, $q = q_v$, etc. In this case, the character ω_s is unramified and only depends on the coset

$$s + \frac{2\pi i}{\log(q)} \mathbb{Z}.$$

Let

$S(F)$ = the space of Schwartz-Bruhat functions on F .

$S(F)'$ = the space of tempered distributions on F , i.e.,

the space of continuous linear functionals $\lambda : S(F) \rightarrow \mathbb{C}$.

Examples.

(i) For F nonarchimedean, a complex valued function f on F is in $S(F)$ if and only if there is an integer $r \geq 0$, depending on f , such that

$$\text{supp}(f) \subset \mathcal{P}^{-r} \quad (\text{compact support})$$

and

$$f = \text{constant on cosets of } \mathcal{P}^r. \quad (\text{locally constant})$$

The space $S(F)$ is simply the complex vector space of all such functions and the tempered distributions are arbitrary $\mathbb{C}$ -linear functionals on $S(F)$.

(ii) For $F = \mathbb{R}$ (resp., $\mathbb{C}$), $S(\mathbb{R})$ is the usual Schwartz space⁷ consisting of complex valued functions which, together with all of their derivatives, are rapidly decreasing, e.g.,

$$f(x) = h(x) e^{-\pi x^2} \quad (\text{resp., } f(x) = h(x, \bar{x}) e^{-2\pi x \bar{x}}),$$

for any polynomial $h \in \mathbb{C}[X]$ (resp., $h \in \mathbb{C}[X, Y]$). This is a Frechét space, and the tempered distributions are the continuous linear functionals on it.

The multiplicative group $F^\times$ acts on $S(F)$ by

$$r(a)f(x) = f(xa),$$

for $x \in F$ and $a \in F^\times$, and on $S(F)'$ by

$$\langle r'(a)\lambda, f \rangle = \langle \lambda, r(a^{-1})f \rangle,$$

where $\langle , \rangle$ denotes the pairing of $S(F)'$ and $S(F)$.

Definition 3.1. For a quasicharacter ω of $F^\times$, let

$$S'(\omega) = \{\lambda \in S(F)' \mid r'(a)\lambda = \omega(a)\lambda\}$$

be the space of ω -eigendistributions.

The space $S'(\omega)$ can be analyzed “geometrically”. Note that there are two orbits for the action of $F^\times$ on the additive group

$$F = \{0\} \cup F^\times.$$

⁷A nice reference for the archimedean theory is the book of Friedlander [4]. For the nonarchimedean case, see [1].

Associated to this we have an inclusion

$$C_c^\infty(F^\times) \hookrightarrow S(F),$$

where $C_c^\infty(F^\times)$ is the space of C^∞ -functions with compact support in $F^\times$. By duality, there is an exact sequence of distributions

$$0 \longrightarrow S(F)'_0 \longrightarrow S(F)' \longrightarrow C_c^\infty(F^\times)' \longrightarrow 0, \quad (3.1)$$

where $S(F)'_0$ is the subspace of distributions supported at 0. This sequence is compatible with the action of $F^\times$. Taking the ω -eigenspaces, we have the sequence

$$0 \longrightarrow S'(\omega)_0 \longrightarrow S'(\omega) \longrightarrow C_c^\infty(F^\times)'(\omega), \quad (3.2)$$

where $S'(\omega)_0$ is the space of ω -eigendistributions supported at 0. First, one has the following simple uniqueness result:

Lemma 3.2. *The space $C_c^\infty(F^\times)'(\omega)$ is one dimensional and is spanned by the distribution $\omega(x) d^\times x$. In particular, for any $\lambda \in S'(\omega)$, there is a complex number c such that*

$$\lambda|_{C_c^\infty(F^\times)} = c \cdot \omega(x) d^\times x,$$

i.e., if $\text{supp}(f) \subset F^\times$ is compact, then

$$\langle \lambda, f \rangle = c \cdot \int_{F^\times} f(x) \omega(x) d^\times x.$$

It remains to determine the space $S(F)'_0$ of tempered distributions supported at 0 and the subspace $S'(\omega)_0$. The delta distribution δ_0 defined by

$$\langle \delta_0, f \rangle = f(0)$$

is obviously $F^\times$ -invariant and supported at 0.

Lemma 3.3.

(i) *If F is nonarchimedean, then*

$$S(F)'_0 = \mathbb{C} \cdot \delta_0 \subset S'(\omega_0),$$

where ω_0 is the trivial character. For $\omega \neq \omega_0$, $S'(\omega)_0 = 0$.

(ii) *If $F = \mathbb{R}$, then,⁸ setting $D = \frac{d}{dx}$,*

$$S(F)'_0 = \bigoplus_{k=0}^{\infty} \mathbb{C} \cdot D^k \delta_0,$$

and

$$D^k \delta_0 \in S'(\omega),$$

⁸In the archimedean case, we give only the $F^\times$ -finite distributions here.

where

$$\omega(x) = x^{-k}.$$

(iii) If $F = \mathbb{C}$, then, setting $D = \frac{\partial}{\partial x}$ and $\bar{D} = \frac{\partial}{\partial \bar{x}}$,

$$S(F)'_0 = \bigoplus_{k,l=0}^{\infty} \mathbb{C} \cdot D^k \bar{D}^l \delta_0.$$

Moreover,

$$D^k \bar{D}^l \delta_0 \in S'(\omega),$$

where

$$\omega(x) = x^{-k} \bar{x}^{-l}.$$

The fundamental local uniqueness result, whose proof we will sketch, is the following.

Theorem 3.4. *For any quasicharacter ω of $F^\times$,*

$$\dim S'(\omega) = 1.$$

Remarks.

(i) When $S'(\omega)_0 = 0$, i.e., when there are no ω -eigendistributions supported at 0, the reasoning above shows that the dimension of $S'(\omega)$ is at most one, so that only existence is at stake. This amounts to showing that the distribution $\omega(x) d^\times x$ in Lemma 3.2 can be extended to an ω -eigendistribution on $S(F)$. This will be established via local zeta integrals below.

(ii) When $S'(\omega)_0 \neq 0$, the problem will be to show that the distribution $\omega(x) d^\times x$ does *not* extend to an ω -eigendistribution. This more delicate fact also comes out of the analysis of local zeta integrals.

Example. Suppose that F is nonarchimedean. If ω is a *ramified* character, then $S'(\omega\omega_s)_0 = 0$, so that

$$\dim S'(\omega\omega_s) \leq 1$$

for all s . If ω is unramified with $\omega(x) = t^{\text{ord}(x)}$, as above, then, by Lemma 3.3,

$$\dim S'(\omega\omega_s) \leq \begin{cases} 2 & \text{if } t \cdot q^{-s} = 1, \text{ and} \\ 1 & \text{otherwise.} \end{cases}$$

Zeta integrals

For a character ω of $F^\times$, i.e., a unitary quasicharacter, the *local zeta integral*

$$z(s, \omega; f) := \int_{F^\times} f(x) \omega\omega_s(x) d^\times x = \int_F f(x) \omega(x) \mu |x|^{s-1} dx, \quad (3.3)$$

is absolutely convergent for all $f \in S(F)$ provided $\text{Re}(s) > 0$. Here $d^\times x = \mu |x|^{-1} dx$, as before. In this range, the distribution $f \mapsto z(s, \omega; f)$ defines a nonzero element

$$z(s, \omega) \in S'(\omega\omega_s).$$

The unramified local theory

Suppose that F is nonarchimedean and suppose that ω is unramified, with $\omega(\varpi) = t$. If f has compact support in $F^\times$, then the integral

$$\int_{F^\times} f(x) \omega \omega_s(x) d^\times x$$

is entire. The idea, then, is to kill the support of an arbitrary $f \in S(F)$ by applying a suitable element of the group algebra $\mathbb{Z}[F^\times]$, more precisely, the element

$$\tau = [1] - [\varpi^{-1}], \quad (3.4)$$

where ϖ is a generator of $\mathcal{P}$. Since any $f \in S(F)$ is constant in a sufficiently small neighborhood, say $\mathcal{P}^r$, of 0, we have, for any $x \in \mathcal{P}^{r+1}$,

$$(r(\tau)f)(x) = f(x) - f(x\varpi^{-1}) = 0.$$

Thus, there is a distribution

$$z_o(s, \omega) \in S'(\omega \omega_s)$$

defined by

$$\langle z_o(s, \omega), f \rangle := \int_{F^\times} (r(\tau)f)(x) \omega \omega_s(x) d^\times x. \quad (3.5)$$

On the other hand, in the halfplane $\operatorname{Re}(s) > 0$, where it is not necessary to kill support to have convergence,

$$\begin{aligned} \langle z_o(s, \omega), f \rangle &= \int_{F^\times} (f(x) - f(\varpi^{-1}x)) \omega \omega_s(x) d^\times x \\ &= \int_{F^\times} f(x) \omega \omega_s(x) d^\times x - \int_{F^\times} f(\varpi^{-1}x) \omega \omega_s(x) d^\times x \\ &= (1 - \omega \omega_s(\varpi)) \cdot z(s, \omega; f) \\ &= (1 - t q^{-s}) \cdot z(s, \omega; f) \\ &= L(s, \omega)^{-1} \cdot \langle z(s, \omega), f \rangle. \end{aligned}$$

Strictly in terms of distributions, this says that, for $\operatorname{Re}(s) > 0$,

$$z(s, \omega) = L(s, \omega) \cdot z_o(s, \omega), \quad (3.6)$$

and, hence, the expression on the right-hand side gives the meromorphic analytic continuation of $z(s, \omega)$ to the whole s plane. Moreover, (3.6) provides an interpretation of the local L -factor as a constant of proportionality between natural bases for the one-dimensional space $S'(\omega \omega_s)$, away from the poles of $L(s, \omega)$.

Finally, to see why the distribution $z_o(s, \omega)$ is “natural,” we normalize the multiplicative Haar measure $d^\times x$ so that the units $\mathcal{O}^\times$ have volume 1. Then, we let

$$f^o = \text{the characteristic function of } \mathcal{O}, \quad (3.7)$$

the ring of integers in F , and compute:

$$\begin{aligned} \langle z_o(s, \omega), f^o \rangle &= \int_{F^\times} (f^o(x) - f^o(\varpi^{-1}x)) \omega \omega_s(x) d^\times x \\ &= \int_{\mathcal{O}^\times} d^\times x \\ &= 1, \end{aligned} \quad (3.8)$$

for all s . The point is that $r(\tau)f^o$ is the characteristic function of $\mathcal{O}^\times$. Thus, $z_o(s, \omega)$ is never zero and gives a basis vector for $S'(\omega \omega_s)$ for all s .

Remark (the L -factor as greatest common denominator). Relations (3.6) and (3.8) together show that $L(s, \omega)$ gives precisely the poles of the family of zeta integrals (3.3). More precisely, for any $f \in S(F)$, the function

$$\frac{z(s, \omega; f)}{L(s, \omega)} = \langle z_o(s, \omega), f \rangle \quad (3.9)$$

is entire and that, for any given s , there is an f (specifically f^o) for which the value of (3.9) at s is nonzero.

Example. To complete the proof of Theorem 3.4 in the unramified case, there is still one point to check, namely that the space $S'(\omega_0)$ of invariant distributions, which contains $z_o(0, \omega_0)$ and has dimension at most 2, is actually only one dimensional. Here is the argument. There is an exact sequence of distributions

$$0 \longrightarrow \mathbb{C} \cdot \delta_0 \longrightarrow S(F)' \longrightarrow C_c^\infty(F^\times)' \longrightarrow 0 \quad (3.10)$$

and, passing to ω_0 -eigendistributions, i.e., $F^\times$ -invariant distributions, we have

$$0 \longrightarrow \mathbb{C} \cdot \delta_0 \longrightarrow S'(\omega_0) \longrightarrow \mathbb{C} \cdot d^\times x. \quad (3.11)$$

The point now is to show that, although $d^\times x$ is in the image of the quotient map in (3.10), it is not in the image of the quotient map in (3.11), i.e., the preimage of $d^\times x$ in $S(F)'$ is *not* invariant.

Explicitly, the distribution $\lambda_0 \in S(F)'$ defined by

$$\langle \lambda_0, f \rangle := \langle d^\times x, f - f(0)f^o \rangle \quad (3.12)$$

gives a preimage of $d^\times x$ in $S(F)'$, which is clearly invariant under $\mathcal{O}^\times$ (since f^o is) and satisfies $\langle \lambda_0, f^o \rangle = 0$. The distributions $r'(\varpi)\lambda_0$ and λ_0 agree on $C_c^\infty(F^\times)$ and hence differ by a distribution supported at 0, i.e.,

$$r'(\varpi)\lambda_0 = \lambda_0 + c \delta_0,$$

for some constant c . To determine c (which would be 0 if λ_0 were actually $F^\times$ -invariant), we evaluate on f^o , using the group algebra element τ , as above:

$$\begin{aligned}
 c &= \langle r'(\varpi)\lambda_0 - \lambda_0, f^o \rangle \\
 &= -\langle \lambda_0, r(\tau)f^o \rangle \\
 &= -\langle d^\times x, r(\tau)f^o \rangle \\
 &= -\langle z_o(0, \omega_0), f^o \rangle \\
 &= -1.
 \end{aligned} \tag{3.13}$$

Thus, on the two-dimensional subspace of $S(F)'$ spanned by λ_0 and δ_0 , $F^\times$ acts by the representation

$$\rho(x) = \begin{pmatrix} 1 & -\text{ord}(x) \\ & 1 \end{pmatrix}. \tag{3.14}$$

This finishes the proof of Theorem 3.4 in the unramified case.

The ramified local theory

Suppose that F is nonarchimedean and that ω is ramified, hence nontrivial on $\mathcal{O}^\times$. The conductor $c(\omega)$ of ω is the smallest integer c such that ω is trivial on $1 + \mathcal{P}^c$. Since any $f \in S(F)$ is constant in a neighborhood of 0, the integral

$$\int_{F^\times - \mathcal{P}^n} f(x) \omega \omega_s(x) d^\times x$$

is independent of n for n sufficiently large. This gives the entire analytic continuation of $z(s, \omega; f)$ to the whole s plane and hence gives a basis vector

$$z_o(s, \omega) := z(s, \omega), \tag{3.15}$$

for the one-dimensional space $S'(\omega \omega_s)$ for all s . This finishes the proof of Theorem 3.4 in the ramified case. We set

$$L_v(s, \omega) = 1 \tag{3.16}$$

in this case, and we note that, for

$$f^o(x) = \begin{cases} \omega(x)^{-1} & \text{if } x \in \mathcal{O}^\times, \text{ and} \\ 0 & \text{otherwise,} \end{cases} \tag{3.17}$$

we have

$$\langle z_o(s, \omega), f^o \rangle = 1.$$

Thus the gcd property (3.9) again holds.

The archimedean case

In this case, it is the whole Taylor series at 0 of the function f which accounts for the poles of the zeta integral $z(s, \omega; f)$, rather than just the value $f(0)$ which arose in the nonarchimedean case; cf. [2, Proposition 3.1.7, p. 271]. It is easy to obtain the analytic continuation of the zeta integral $z(s, \omega; f)$ by integration by parts.

Here we just summarize the results, using the normalizations of [13]. The convention that ω is unitary is dropped in order to obtain particularly concise expressions.

If $F = \mathbb{R}$, any quasicharacter has the form $\omega\omega_s$ where $s \in \mathbb{C}$ and

$$\omega(x) = x^{-a}, \quad a = 0, 1. \quad (3.18)$$

Let

$$L(s, \omega) = \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right). \quad (3.19)$$

If $F = \mathbb{C}$, any quasicharacter has the form $\omega\omega_s$ where $s \in \mathbb{C}$ and

$$\omega(x) = x^{-a} \bar{x}^{-b}, \quad a, b \in \mathbb{Z}, \quad \min(a, b) = 0. \quad (3.20)$$

Let

$$L(s, \omega) = (2\pi)^{1-s} \Gamma(s). \quad (3.21)$$

Proposition 3.5.

(i) *The distribution*

$$z_o(s, \omega) := L(s, \omega)^{-1} z(s, \omega)$$

has an entire analytic continuation to the whole s plane and, for all s , defines a basis vector for the space $S'(\omega\omega_s)$.

(ii) *If*

$$f^o(x) = \begin{cases} f_a(x) := x^a e^{-\pi x^2} & \text{if } F = \mathbb{R}, \text{ and} \\ f_{a,b}(x) := x^a \bar{x}^b e^{-2\pi x \bar{x}} & \text{if } F = \mathbb{C}, \end{cases}$$

then

$$\langle z_o(s, \omega), f^o \rangle = 1.$$

Note that we are using the measure $d^\times x = |x|^{-1} dx$, where $|x|$ is the usual absolute value (resp., $|x| = x\bar{x}$) and dx is Lebesgue measure (resp., twice Lebesgue measure) when $F = \mathbb{R}$ (resp., $F = \mathbb{C}$). For (ii), see [13, Lemma 8, p. 127].

Remark. In effect, the zeta distribution $z(s, \omega)$ has a meromorphic analytic continuation in s with simple poles at certain nonpositive integers. For example, when $F = \mathbb{R}$, the poles occur at the points $-r \in \mathbb{Z}_{\leq 0}$ with $r \equiv a \pmod{2}$, and the residues are

$$\operatorname{Res}_{s=-r} z(s, \omega) = c_r D^r \delta_0, \quad (3.22)$$

with a nonzero constant c_r . Here D is as in Lemma 3.3. The constant term in the Laurent expansion of $z(s, \omega)$ at $s = -r$ gives an extension to $S(F)$ of the distribution $x^{-r} d^\times x$ on $C_c^\infty(F^\times)$. A simple calculation, like (3.9), shows that this extension does *not* lie in the space $S'(\omega\omega_{-r})$. In the case $F = \mathbb{C}$, the residue at $s = -r$ lies in $S'(\omega\omega_{-r})$, where $\omega\omega_{-r}(x) = x^{-a-r} \bar{x}^{-b-r}$ and is a nonzero multiple of $D^{a+r} \bar{D}^{b+r} \delta_0$.

Fourier transforms

Fix a nontrivial character ψ of the additive group $F = F^+$, and identify F with its topological dual character group

$$\hat{F} := \operatorname{Hom}_{\text{cont}}(F, \mathbb{C}^1),$$

by

$$F \xrightarrow{\sim} \hat{F}, \quad y \mapsto (x \mapsto \psi(xy)).$$

For F nonarchimedean, the conductor $\nu(\psi)$ of ψ is the largest integer ν such that ψ is trivial on $\mathcal{P}^{-\nu}$.

The Fourier transform

$$\hat{f}(x) = \int_F f(y) \psi(xy) dy$$

of a function $f \in S(F)$ is well defined and again lies in $S(F)$. (This is a fundamental property of the space of Schwartz–Bruhat functions.) The map $f \mapsto \hat{f}$ is an isomorphism $\hat{\cdot}: S(F) \xrightarrow{\sim} S(F)$. There is a unique choice of the Haar measure, the self-dual measure with respect to ψ , such that Fourier inversion gives

$$\hat{\hat{f}}(x) = f(-x).$$

Supposing ψ to be given, we fix this choice of dx from now on. The Fourier transform of a distribution is defined by

$$\langle \hat{\lambda}, f \rangle = \langle \lambda, \hat{f} \rangle.$$

Lemma 3.6. *If $\lambda \in S'(\omega)$ is an ω -eigendistribution, then*

$$\hat{\lambda} \in S'(\omega^{-1}\omega_1),$$

i.e., $\hat{\lambda}$ is an $\omega^{-1}\omega_1$ -eigendistribution, where $\omega_1(x) = |x|$.

Proof. An exercise! □

By the uniqueness result, Theorem 3.4, it follows that $\widehat{z_o(s, \omega)}$ is a constant multiple (depending on s and on ψ) of $z_o(1-s, \omega^{-1})$.

Corollary 3.7 (the local functional equation).

$$z_o(\widehat{1-s}, \omega^{-1}) = \epsilon(s, \omega, \psi) z_o(s, \omega) \quad (3.23)$$

for a nonzero constant $\epsilon(s, \omega, \psi)$, the local epsilon factor, depending on s , ω , and ψ .

Remark. For a given $f \in S(F)$, the local functional equation can be written as the relation

$$\frac{z(1-s, \omega^{-1}; \hat{f})}{L(1-s, \omega^{-1})} = \epsilon(s, \omega, \psi) \cdot \frac{z(s, \omega; f)}{L(s, \omega)}, \quad (3.24)$$

or, alternatively, as

$$z(1-s, \omega^{-1}; \hat{f}) = \epsilon(s, \omega, \psi) \frac{L(1-s, \omega^{-1})}{L(s, \omega)} \cdot z(s, \omega; f), \quad (3.25)$$

where the constant of proportionality

$$\gamma(s, \omega, \psi) = \epsilon(s, \omega, \psi) \frac{L(1-s, \omega^{-1})}{L(s, \omega)}, \quad (3.26)$$

which can now have zeroes and poles, is called the local gamma factor. This is the more traditional formulation, while that in Corollary 3.7 is emphasized in [12].

Various important properties of the ϵ -factor $\epsilon(s, \omega, \psi)$ can be obtained from relation (3.23). For example, if f^o is the standard function with respect to ω , then, evaluating both sides of (3.23) on f^o yields the useful identity

$$\epsilon(s, \omega, \psi) = \langle z_o(1-s, \omega^{-1}), \widehat{f^o} \rangle. \quad (3.27)$$

By construction, the zeta distribution satisfies

$$z(s, \omega\omega_t) = z(s+t, \omega).$$

Since the local L -function is also required to have this property, we have

$$z_o(s, \omega\omega_t) = z_o(s+t, \omega)$$

as well. The standard function f^o does not change under multiplication by ω_t . This implies that

$$\epsilon(s, \omega\omega_t, \psi) = \epsilon(s+t, \omega, \psi). \quad (3.28)$$

For $\beta \in F^\times$, let $\psi_\beta(x) = \psi(\beta x)$. If dx is the self-dual measure with respect to ψ , then $|\beta|^{\frac{1}{2}} dx$ is the self-dual measure with respect to ψ_β . If $\hat{f}$ is the Fourier transform of $f \in S(F)$ with respect to ψ , then $|\beta|^{\frac{1}{2}} r(\beta) \hat{f}$ is the Fourier transform with respect to ψ_β as always for the self-dual measure. Thus, inserting this on the right side of (3.27), we obtain

$$\epsilon(s, \omega, \psi_\beta) = |\beta|^{s-\frac{1}{2}} \omega(\beta) \epsilon(s, \omega, \psi). \quad (3.29)$$

The following result gives explicit values for the ϵ -factors. Using the two relations (3.28) and (3.29), it suffices to compute these factors with both ψ and ω in “standard” form.

Recall that, in the nonarchimedean case, $\nu = \nu(\psi)$ is the conductor of ψ . In the archimedean case, the “standard” choice

$$\psi(x) = \begin{cases} e(x) & \text{if } F = \mathbb{R}, \\ e(x + \bar{x}) & \text{if } F = \mathbb{C}, \end{cases} \quad (3.30)$$

where, for a real number t , $e(t) = e^{2\pi it}$, has self-dual measures the measures described after Proposition 3.5.

Proposition 3.8.

(i) For F nonarchimedean, ω unramified, and $\nu(\psi) = \nu$:

$$\epsilon(s, \omega, \psi) = \omega(\varpi^\nu) q^{(\frac{1}{2}-s)\nu}.$$

(ii) For F nonarchimedean, ω ramified with conductor c , and $\nu(\psi) = \nu$,

$$\epsilon(s, \omega, \psi) = \omega(\varpi^{\nu+c}) q^{(\frac{1}{2}-s)(\nu+c)} \mathfrak{g}(\omega, \psi),$$

where $\mathfrak{g}(\omega, \psi)$ is the Gauss sum:

$$\mathfrak{g}(\omega, \psi) = q^{\frac{1}{2}(\nu+c)} \int_{\mathcal{O}^\times} \omega^{-1}(y) \psi(\varpi^{-\nu-c} y) dy.$$

with dy the self-dual measure on F with respect to ψ .

(iii) For $F = \mathbb{R}$, write $\omega(x) = x^{-a}$ with $a = 0$, or 1 and let ψ be as in (3.30). Then

$$\epsilon(s, \omega, \psi) = i^a.$$

(iv) For $F = \mathbb{C}$, write $\omega(x) = x^{-a} \bar{x}^{-b}$, for $a, b \in \mathbb{Z}$ with $\min(a, b) = 0$ and let ψ be as in (3.30). Then

$$\epsilon(s, \omega, \psi) = i^{\max(a, b)}.$$

Proof. By (3.27), one has to compute the Fourier transform of the standard function f° for ω , where the character ψ can be taken to be standard. For example, in case (i), f° is the characteristic function of $\mathcal{O}$, and, assuming that $\nu(\psi) = 0$, $\widehat{f^\circ} = f^\circ$. This gives $\epsilon(s, \omega, \psi) = 1$ in this case. The general unramified case is then immediate via (3.29).

In case (ii), f° is given by (3.17). Then, following [13, Proposition 13, p. 131],

$$\widehat{f^\circ}(x) = \int_{\mathcal{O}^\times} \omega^{-1}(y) \psi(xy) dy$$

vanishes unless $\text{ord}(x) = -\nu - c$ where $c = c(\omega)$ is the conductor of ω . In fact,

$$\widehat{f^\circ}(x) = \mathfrak{g}(\omega, \psi) q^{-\frac{1}{2}(\nu+c)} \overline{f^\circ(\varpi^{\nu+c} x)}, \quad (3.31)$$

where

$$\mathfrak{g}(\omega, \psi) = q^{\frac{1}{2}(\nu+c)} \int_{\mathcal{O}^\times} \omega^{-1}(y) \psi(\varpi^{-\nu-c} y) dy. \quad (3.32)$$

Applying (3.32) twice yields $\mathfrak{g}(\omega, \psi) \overline{\mathfrak{g}(\omega, \psi)} = 1$. Since $\overline{f^o}$ is the standard function for ω^{-1} , we obtain

$$\epsilon(s, \omega, \psi) = \omega(\varpi^{\nu+c}) \mathfrak{g}(\omega, \psi) q^{(\frac{1}{2}-s)(\nu+c)}.$$

In case (iii), we compute, for $f^o = f_a$, as in Proposition 3.5,

$$\begin{aligned} \widehat{f_a}(x) &= \int_{\mathbb{R}} y^a e^{-\pi y^2} e(xy) dy \\ &= (2\pi i)^{-a} D_x^a \left\{ \int_{\mathbb{R}} e^{-\pi y^2} e(xy) dy \right\} \\ &= (2\pi i)^{-a} D_x^a \{ e^{-\pi x^2} \} \\ &= i^a f_a(x). \end{aligned} \quad (3.33)$$

Since f_a is also standard for $\omega^{-1} = \omega \omega_2$, we obtain (iii).

In case (iv), we have $f^o = f_{a,b}$, as in Proposition 3.5. Then, recalling that $\min(a, b) = 0$,

$$\begin{aligned} \widehat{f_{a,b}}(x) &= \int_{\mathbb{C}} y^a \bar{y}^b e^{-2\pi y \bar{y}} e(xy + \bar{x} \bar{y}) dy \\ &= (2\pi i)^{-a} D_x^a (2\pi i)^{-b} \bar{D}_x^b \left\{ \int_{\mathbb{C}} e^{-2\pi y \bar{y}} e(xy + \bar{x} \bar{y}) dy \right\} \\ &= (2\pi i)^{-a} D_x^a (2\pi i)^{-b} \bar{D}_x^b \{ e^{-2\pi x \bar{x}} \} \\ &= i^{a+b} f_{b,a}(x). \end{aligned} \quad (3.34)$$

Noting that $f_{b,a}$ is the standard function for $\omega^{-1}(x) = \bar{x}^{-a} x^{-b} |x|^{a+b}$, we obtain the claimed value. More details can be found in Tate's thesis or in [13]. $\square$

4 Global theory

Distributions

Returning to the number field k with its adèle ring $\mathbb{A}$, we define the space $S(\mathbb{A})$ of Schwartz–Bruhat functions on $\mathbb{A}$. This space contains all functions of the form

$$f = \otimes_v f_v,$$

where $f_v \in S(k_v)$ for all v and $f_v = f_v^o$, the characteristic function of $\mathcal{O}_v$, for almost all v . Note that

$$f(x) = \prod_v f_v(x_v)$$

is then well defined, as almost all factors are 1. The finite linear combinations of such factorizable functions are dense in $S(\mathbb{A})$. Again, $S(\mathbb{A})'$ is the space of tempered distributions, i.e., continuous linear functionals on $S(\mathbb{A})$.

Lemma 4.1. *Suppose that $\{\lambda_v\}$ is a collection of local tempered distributions $\lambda_v \in S(k_v)'$ such that, for almost all v , $\langle \lambda_v, f_v^o \rangle = 1$. Then there is a unique tempered distribution $\lambda = \otimes_v \lambda_v$ on $S(\mathbb{A})$, defined on factorizable functions by*

$$\langle \lambda, f \rangle = \prod_v \langle \lambda_v, f_v \rangle.$$

Conversely, if $\lambda \in S(\mathbb{A})'$ is nonzero, choose a factorizable f such that $\lambda(f) = 1$. Then, for any place v , we write

$$f = f_v \otimes f^v, \quad f^v = \otimes_{w \neq v} f_w,$$

and we define a map

$$S(k_v) \longrightarrow S(\mathbb{A}), \quad g_v \mapsto g_v \otimes f^v,$$

by just varying the v component. Thus, we obtain local distributions

$$\langle \lambda_v, g_v \rangle := \langle \lambda, g_v \otimes f^v \rangle,$$

with $\langle \lambda_v, f_v^o \rangle = 1$ for almost all v , and $\lambda = \otimes_v \lambda_v$.

The idèle group $\mathbb{A}^\times$ acts on $S(\mathbb{A})$ and $S(\mathbb{A})'$, and, for a quasicharacter ω of $\mathbb{A}^\times$, trivial on $k^\times$, we let $S'(\omega)$ be the space of ω -eigendistributions. The factorization $\omega = \otimes_v \omega_v$ of the global quasicharacter gives rise to the decomposition

$$S'(\omega) = \otimes_v S'_v(\omega_v),$$

where we write $S'_v = S(k_v)'$. The local uniqueness Theorem 3.4 then yields the global uniqueness result:

Theorem 4.2. *For all quasicharacter ω of $\mathbb{A}^\times$, the space of global ω -eigendistributions $S'(\omega)$ has dimension 1. For any $s \in \mathbb{C}$, the space $S'(\omega \omega_s)$ is spanned by the standard $\omega \omega_s$ -eigendistribution*

$$z_o(s, \omega) := \otimes_v z_o(s, \omega_v).$$

Global zeta integrals and the completed L-function

For a character ω of $k^\times \backslash \mathbb{A}^\times$ and a function $f \in S(\mathbb{A})$, we can define a global zeta integral by

$$z(s, \omega; f) = \int_{\mathbb{A}^\times} f(x) \omega \omega_s(x) d^\times x. \quad (4.1)$$

If f is factorizable, this is a product of the corresponding local integrals

$$z(s, \omega; f) = \prod_v z(s, \omega_v; f_v). \quad (4.2)$$

A key point here is that this integral *converges* precisely when the product of the local integrals converges. But, since there is a finite set of places S such that for $v \notin S$, ω_v is unramified and $f_v = f_v^o$, we obtain convergence whenever the product

$$L^S(s, \omega) = \prod_{v \notin S} L_v(s, \omega_v)$$

is absolutely convergent, i.e., in the half plane $\operatorname{Re}(s) > 1$.

In terms of distributions, in the half plane $\operatorname{Re}(s) > 1$, we have $z(s, \omega) = \otimes_v z(s, \omega_v)$, and

$$z(s, \omega) = \Lambda(s, \omega) \cdot z_o(s, \omega), \quad (4.3)$$

where we define the *complete L-function*

$$\Lambda(s, \omega) := \prod_v L_v(s, \omega), \quad (4.4)$$

now including all local L -factors, and

$$z_o(s, \omega) = \otimes_v z_o(s, \omega_v), \quad (4.5)$$

is the standard $\omega\omega_s$ -eigendistribution defined above. Thus, in the half-plane of convergence, the complete L -function can be viewed as a factor of proportionality between two “natural” global eigendistributions.

Fourier transforms and the global functional equation

We fix a global additive character ψ of $\mathbb{A}$, trivial on k . By restricting to local components, we can write

$$\psi(x) = \prod_v \psi_v(x_v).$$

For each nonarchimedean place v , the conductor $v_v := v(\psi_v)$ of ψ_v is defined. We then get an identification of $\mathbb{A}$ with its continuous character group, compatible with those defined by the components ψ_v in the local cases. We define a global Fourier transform

$$\hat{} : S(\mathbb{A}) \xrightarrow{\sim} S(\mathbb{A}),$$

compatible with the local ones, i.e., if f is factorizable, then

$$\hat{f} = \otimes_v \hat{f}_v.$$

The ψ -self dual measure dx on $\mathbb{A}$ can be written as a product of the ψ_v -self dual measures on the k_v 's. We define the Fourier transform of a distribution as before.

The desired analytic continuation and functional equation of the complete L -function $\Lambda(s, \omega)$ will follow from the conjunction of (i) the proportionality (4.3), (ii) the local functional equations of the $z_o(s, \omega_v)$'s and (iii) the following global identity:

Theorem 4.3. *The distribution $z(s, \omega)$, defined by the global zeta integral (4.1) for $\operatorname{Re}(s) > 1$ has a meromorphic analytic continuation to the whole s plane and satisfies the functional equation*

$$z(\widehat{1 - s, \omega^{-1}}) = z(s, \omega).$$

Assuming Theorem 4.3, we immediately obtain the corresponding analytic continuation of $\Lambda(s, \omega)$ from (4.3), since the “normalized” global distribution $z_o(s, \omega)$ is entire. Recall that the local functional equations have the form (3.13):

$$z_o(\widehat{1 - s, \omega_v^{-1}}) = \epsilon_v(s, \omega_v, \psi_v) z_o(s, \omega_v),$$

where $\epsilon_v(s, \omega_v, \psi_v)$ is given explicitly in Proposition 3.7. Note that almost all of the $\epsilon_v(s, \omega_v, \psi_v)$ ’s are 1, so we can define the *global epsilon factor*

$$\epsilon(s, \omega) = \prod_v \epsilon_v(s, \omega_v, \psi_v), \quad (4.6)$$

and obtain the functional equation

$$z_o(\widehat{1 - s, \omega^{-1}}) = \epsilon(s, \omega) z_o(s, \omega), \quad (4.7)$$

by taking products of local ones. Here the additive character ψ has been omitted from the notation since, in fact, the product no longer depends on the choice made! This is due to the relations (3.29) for all places v , the fact that every global additive character has the form ψ_β , for $\beta \in k^\times$ for a given nontrivial ψ , and the product formula. Then,

$$\begin{aligned} \Lambda(1 - s, \omega^{-1}) \cdot z_o(\widehat{1 - s, \omega^{-1}}) &= z(\widehat{1 - s, \omega^{-1}}) \\ &= z(s, \omega) \\ &= \Lambda(s, \omega) \cdot z_o(s, \omega). \end{aligned} \quad (4.8)$$

Substituting relation (4.7) in the first expression in (4.8) yields

$$\Lambda(1 - s, \omega^{-1}) \cdot \epsilon(s, \omega) \cdot z_o(s, \omega) = \Lambda(s, \omega) \cdot z_o(s, \omega). \quad (4.9)$$

But, finally, the distribution $z_o(s, \omega)$ is nowhere zero by construction, and so we obtain *the functional equation*:

Corollary 4.4.

$$\Lambda(s, \omega) = \epsilon(s, \omega) \cdot \Lambda(1 - s, \omega^{-1}).$$

Finally, the proof of Theorem 4.3 comes down to Riemann’s classic argument based on Poisson summation! For this very standard calculation, we refer the reader to Tate’s thesis, pp. 339–341, Weil [13, pp. 121–124], or Bump [2, pp. 267–270].

5 Examples

Here we will be very brief.

For a number field k and a (unitary) character $\omega = \otimes_v \omega_v$ of $\mathbb{A}^\times/k^\times$, let

$$\mathfrak{m} = \prod_v \mathfrak{p}_v^{c_v(\omega)}, \quad (5.1)$$

where the product is taken over nonarchimedean places and $c_v(\omega) = c_v(\omega_v)$ is the conductor of ω_v . Thus, for a finite place v , $\mathfrak{p}_v \nmid \mathfrak{m}$ if and only if ω is unramified at v . We can define a character χ of $I_{\mathfrak{m}}$, the group of fractional ideals of k prime to $\mathfrak{m}$ by putting $\chi(\mathfrak{p}_v) = \omega_v(\varpi_v)$. Since ω_v is unramified, this value is independent of the choice of local generator ϖ_v . Let

$$k_{\mathfrak{m}}^\times = \{\alpha \in k^\times \mid \alpha \equiv 1 \pmod{(\mathfrak{m})}\}, \quad (5.2)$$

where the condition $\alpha \equiv 1 \pmod{(\mathfrak{m})}$ means that $\text{ord}_v(\alpha - 1) \geq c_v(\omega)$ for all v with $c_v(\omega) > 0$ and no condition is imposed on α at other places. Let $P_{\mathfrak{m}}$ denote the group of principal fractional ideals (α) generated by elements $\alpha \in k_{\mathfrak{m}}^\times$. The fact that $\omega(\alpha) = 1$ then implies that

$$\chi((\alpha)) = \omega_\infty(\alpha)^{-1}, \quad (5.3)$$

where (α) is the principal ideal generated by α and $\omega_\infty = \prod_{v \text{ arch.}} \omega_v$ is the archimedean part of ω . Thus, a character ω determines a character χ of $I_{\mathfrak{m}}$ whose restriction to $P_{\mathfrak{m}}$ is given by evaluation of an archimedean character on a generator of the ideal.⁹ Since $P_{\mathfrak{m}}$ has finite index in $I_{\mathfrak{m}}$ there are a finite number of such characters with a given conductor and a given archimedean part. For a fixed $\mathfrak{m}$, this archimedean character is constrained by the requirement that if $\epsilon \in k_{\mathfrak{m}}^\times \cap \mathcal{O}_k^\times$ is a unit with $\epsilon \equiv 1 \pmod{(\mathfrak{m})}$, then

$$\omega_\infty(\epsilon) = 1. \quad (5.4)$$

Conversely, any ω_∞ satisfying this condition for some $\mathfrak{m}$ determines a character of $P_{\mathfrak{m}}$ which can be extended to give a finite collection of Grössencharacters χ of $I_{\mathfrak{m}}$.

Of course, one can recover ω from a classical Hecke character χ . Starting from such a χ with given archimedean type ω_∞ , i.e., satisfying (5.3), we define unramified characters ω_v of $k_v^\times$ for $\mathfrak{p}_v \nmid \mathfrak{m}$ by setting $\omega_v(\varpi_v) = \chi(\mathfrak{p}_v)$. The remaining components ω_v of ω for v with $\mathfrak{p}_v \mid \mathfrak{m}$ are then determined uniquely by the condition that $\omega(\alpha) = 1$ for all $\alpha \in k^\times$. More precisely, the weak approximation theorem [3] implies that the image of $k^\times$ is dense in the product $\prod_{v \in S} k_v^\times$ for any finite set of places S . The component ω_v has conductor at most $\text{ord}_v(\mathfrak{m})$ and conductor precisely $\text{ord}_v(\mathfrak{m})$ if χ is primitive modulo $\mathfrak{m}$, and so on.

⁹I believe that this is what Hecke means by the term Grössencharakter, i.e., that on $(\alpha) \in P_{\mathfrak{m}}$, χ depends on the quantities (Grössen) $\alpha_v \in k_v$, for v archimedean.

For $\omega_\infty = 1$, we may take any $\mathfrak{m}$. For $\mathfrak{m} = \mathcal{O}_k$, we obtain everywhere unramified characters, i.e., characters of the ideal class group $H = I/P$. For more general $\mathfrak{m}$, we obtain characters of the ray class group $H_{\mathfrak{m}} = I_{\mathfrak{m}}/P_{\mathfrak{m}}$ mentioned in Kowalski's chapters.

The case $\omega = 1$ gives, of course, the Dedekind zeta function of the field k . This example is explained in detail in [13, Section 6, of Chapter VII] while the case of a more general ω is explained in Section 7 of that chapter.

Exercise. Let k be the real quadratic field $k = \mathbb{Q}(\sqrt{5})$. Then $\mathcal{O}_k = \mathbb{Z} + \mathbb{Z}\epsilon$, where $\epsilon = \frac{1+\sqrt{5}}{2}$ is the fundamental unit, and $\mathcal{O}_k^\times = \{\pm 1\} \times \{\epsilon^n \mid n \in \mathbb{Z}\}$.

(i) First suppose that $\mathfrak{m} = \mathcal{O}_k$. Then the character

$$\omega_\infty(x_1, x_2) = \left| \frac{x_1}{x_2} \right|^{-i\pi/\log \epsilon}$$

is trivial on $\mathcal{O}_k^\times$, and hence defines a Hecke character χ of I by

$$\chi((\alpha)) = \omega_\infty(\alpha)^{-1} = \left| \frac{\alpha}{\alpha'} \right|^{i\pi/\log \epsilon}.$$

Write down the corresponding Hecke L -series and its functional equation.

Next take $\mathfrak{m} = (\sqrt{5}) = \mathfrak{p}_5$.

(ii) Check that the group $k_{\mathfrak{m}}^\times \cap \mathcal{O}_k^\times$ of units $u \equiv 1 \pmod{(\mathfrak{m})}$ is generated by $-\epsilon^2$.

(iii) Determine all characters $\omega_\infty : \mathbb{R}^\times \times \mathbb{R}^\times \rightarrow \mathbb{C}^\times$ which satisfy (5.3) for $\mathfrak{m}$. For example,

$$\omega_\infty(x_1, x_2) = \text{sgn}(x_1) \left| \frac{x_1}{x_2} \right|^{-i\pi/(4 \log \epsilon)} \quad \text{and} \quad \omega_\infty(x_1, x_2) = \left| \frac{x_1}{x_2} \right|^{-i\pi/(2 \log \epsilon)} \quad (5.4)$$

are two possibilities.

(iv) For a given ω_∞ , determine the local component ω_5 for $\mathfrak{p}_5 = \mathfrak{m}$. Note that $\mathcal{O}_5^\times / (1 + \mathcal{P}_5) \simeq \mathbb{F}_5^\times$ and that the global units map onto this group. Also, $\varpi_5 = \sqrt{5}$ gives a local uniformizer. For example, for ω_∞ given by (5.4), ω_5 is the ramified character with $\omega_5(\sqrt{5}) = 1$ and with $\omega_5(\epsilon) = i$.

(v) Determine the classical Hecke character χ of $I_{\mathfrak{m}}$ associated to ω .

(vi) Write down the associated Hecke L -series and its functional equation.

REFERENCES

- [1] J. Bernstein and A. Zelevinski, Representations of the group $GL(n, F)$ where F is a nonarchimedean local field, *Russian Math. Surveys*, **3**(1976), 1–68.
- [2] D. Bump, *Automorphic Forms and Representations*, Cambridge University Press, Cambridge, UK, 1997.

- [3] J. W. S. Cassels and A. Fröhlich, eds., *Algebraic Number Theory*, Thompson Book Company, Washington, DC, 1967.
- [4] F. G. Friedlander, *Introduction to the Theory of Distributions*, Cambridge University Press, Cambridge, UK, 1982.
- [5] S. Gelbart, *Automorphic Forms on Adele Groups*, Princeton University Press, Princeton, NJ, 1975.
- [6] S. Gelbart, An elementary introduction to the Langlands program, *Bull. Amer. Math. Soc.*, **10**(1984), 177–219.
- [7] E. Hecke, Eine neue Art von Zetafunktionen und ihre Beziehungen zur Verteilung der Primzahlen, Erste und Zweite Mitteilung, in *Mathematische Werke*, Vandenhoeck and Ruprecht, Göttingen, The Netherlands, 1959.
- [8] E. Kowalski, Elementary theory of L -functions I and II, in J. Bernstein and S. Gelbart, eds., *An Introduction to the Langlands Program*, Birkhäuser Boston, Boston, 2003, 1–37 (this volume).
- [9] S. Lang, *Algebraic Number Theory*, Addison–Wesley, Reading, MA, 1970.
- [10] J. Tate, *Fourier Analysis in Number Fields and Hecke's Zeta-Functions*, Ph.D. thesis, Princeton University, Princeton, NJ, 1950.
- [11] R. Valenza and D. Ramakrishnan, *Fourier Analysis on Number Fields*, Graduate Texts in Mathematics 186, Springer-Verlag, New York, 1999.
- [12] A. Weil, Fonction zeta et distributions, *Séminar Bourbaki*, **III**(1966); also appears in [14].
- [13] A. Weil, *Basic Number Theory*, Springer-Verlag, New York, 1967.
- [14] A. Weil, *Collected Papers*, Springer-Verlag, New York, 1979.